

Web App: An Enterprise Database Application

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1 Introduction

1. Based on your ER model design, you need to map your ER model to the database schema.
2. You will develop a web application program that combines SQL statements and a host language, which a GUI will automatically execute. You should develop a basic web app functionality sending queries to the DBMS and showing the results on the website.

2 Target Enterprise Applications

Choose the topic of your ER design homework.

3 Major Requirements

1. Application programming: Required language: JavaScript.
The input will be provided by the user in a web form (combining HTML and JavaScript), validated before execution, and then you will build SQL statements, sending them to the MYSQL (combining user-defined input and your internal SQL tables).
2. Constraints: all your tables must fulfill entity and referential integrity.
3. Data: You should populate reference (lookup, not constantly changing) tables with meaningful information with at least 10 rows. There should be at least 2 tables that require modification (updated frequently, recording usage, payment, and interaction). Depending on the target business application, some tables will be written based on input from the web form, and some data will be ready in tables.
4. Database server: We will use the MySQL database installed on your laptop.
5. Languages and OS.
Since your program will work on the browser in a local machine (e.g., laptop), it is expected it will work well on any OS (you are encouraged to test on a Windows, a Linux desktop). The most common OS for the web client is Windows, but it would be a good idea to test it on other devices like a smartphone.
Programming language: JavaScript, combined with HTML CSS etc..
6. You cannot export or read entire tables row by row, doing processing outside the DBMS. You cannot process data outside the DBMS with the host language low-level processing mechanisms (i.e. locking, text files). The bulk of processing must be done with SQL statements, with high-level control in the web app language (if/while/for).
7. Testing:
Your program will be tested on a browser like Chrome, or Safari. Your program will generate and send SQL statements to the DBMS. The app will open on your local computer (e.g., your laptop) and will connect to the local DBMS by default.

3.1 Programming details

1. Query the table:

Your app is expected to produce useful reports for real users on the target application. Imagine these users need useful information but do not care and cannot figure out how to write SELECT statements.

2. Tables to update(bonus points): Update and insert tables depending on your target application and design. It is expected you will update a few records (never all records in a table).

3. DML: It is recommended you use INSERT and UPDATE. Avoid DELETE.

4. Initial database state:

Populate tables with information as much as possible and leave day-to-day web processing for transactions.

5. GUI: Provide many options to avoid executing code from the command line.

The app should provide a basic interactive menu (OK to mimic some real web site). The app should have a button to cancel or stop anything. Your program must be prepared for failures and errors, and informative diagnosis messages in the GUI.

Extra points :

1. Tables to update(bonus points 10): Update and insert tables depending on your target application and design. It is expected you will update a few records (never all records in a table).
2. If you use some popular Framework such as react to develop your web application(5 points).
3. If you can make your web application also works on an iphone or Androids system(5 points).

4 Grading

Deliverables:

1. App: working web app running locally, opened with index.html. Include a link to your README file and attend the final presentation(below).

Your web application should work for SELECT queries, which means you can show some results from the database.

If your web application can work for both SELECT and modify queries such as INSERT, UPDATE, you will receive bonus points.

2. PDF: call it "teamXY.pdf," where XY is your team id (01,02,...,21,..30). Report in PDF format, including a written summary of the web app, including SQL for the transactions if

any, and SQL for a few important queries (i.e., the most important tables), and HTTP link to a demo video (7 minutes at most). Your report should include a list of web references at the end, including websites you choose as references to build your app.

3. Source files: all source code, diagram files, and any useful document you got from the Internet.

Source code and web app originality: It is expected all source code, databases and GUI will be different even if some source code, some tables, and website ideas are obtained from the Internet.

Submission: all JS/HTML source code and PDF will be uploaded to the Canvas.

Score: A well-designed normalized database with correct and efficient SQL and a well-developed and working app will receive a high score. A beautiful GUI is secondary (do not waste too much time with frames, animation, colors, images, or fonts). Given the freedom in the ER design, tables and GUI there is no specific database design correctness score.